

Editorial requests

Please include a sentence on the theoretical motivation for this work at the beginning of the abstract.

We have added a sentence on the theoretical motivation for our work to the beginning of our abstract.

Please provide a point-by-point response to the comments by Reviewer 4. In addition, in line with Reviewer 4's comments, we ask that you include a deeper discussion of the limitations and tone down claims regarding the practical application of these findings.

We have provided a point-by-point response to Reviewer 4's comments below. The reviewer's comments are italicized, and our responses are in bold. In response to the reviewer's comments, we have added language to our *Discussion* section to better acknowledge the limitations of our framework's practical application in its current form, and highlight what would be required to translate our findings into the sorts of meaningful feedback or interventions the reviewer describes.

Please include a 'Supplementary Information' header at the start of the supplements document.

We have added a "Supplementary Information" header to the beginning of the document.

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If it is acceptable, we would prefer to include both the DOI URL and GitHub URL in our data and code availability statements (similarly to other recently published papers on *Nature Communications's* website). However, if this is not allowable, we consent to the Data Availability statement being edited to read:

All of the data analyzed in this manuscript are archived on Zenodo at <https://doi.org/10.5281/zenodo.17783435>. All Khan Academy content is available for free at <https://www.khanacademy.org>.

and to the Code Availability statement being edited to read:

All of the code for running our experiment and carrying out the analyses is archived on Zenodo at <https://doi.org/10.5281/zenodo.17783435>.

Please ensure the references are in the standard Nature format and follow the sequence: author list, title of paper, name of journal, volume number, initial-final page numbers or article number (year).

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Figure 8c was created solely by the authors and does not include any previously created elements. The code we wrote to generate Figure 8c may be found at https://github.com/ContextLab/efficient-learning-khan/blob/master/code/notebooks/main/7_knowledge-maps.ipynb.

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Please remove the hyperlinks to the figures and tables from the supplementary information file and provide it in the plain text.

We have removed the hyperlinks from the Supplementary Information file.

Reviewer comments

Reviewer #4 (Remarks to the Author):

The manuscript by Fitzpatrick and colleagues describes an application of text embedding models to measure and “map” conceptual knowledge during and after learning about some topics in physics (watching videos and then taking quizzes). I did not review prior versions of this manuscript but have been asked to assess the responses of the authors to prior rounds of feedback. Given this context, below I provide my own evaluation of the manuscript (mostly focused on the responses to Reviewer 3’s concerns), but I do not specifically suggest or request any further edits to the manuscript.

First, it is clear that the authors have put a tremendous amount of work into responding to prior comments from reviewers. I think they have been about as responsive as possible and the paper is very well written. In many respects it is a beautiful paper. Reviewers 1 and 2 seem generally satisfied with the revisions. Reviewer 3, however, raised a number of concerns related to details of the statistical analyses, the specific models used, and the overall significance of the findings. My impression is that the authors have satisfactorily addressed Reviewer 3’s comments related to the statistical analyses and have appropriately contextualized the choice of model (with helpful comments about other models). However, I do share some of Reviewer 3’s concerns about the significance of the work—and these concerns are, I think, less addressable unless the scope of the paper was substantially expanded (which I think is infeasible). Thus, I have a mixed impression: the authors have done some very commendable work on this paper and there is certainly a wow factor to the methods and visualizations (which may be enough for the paper to have impact), but I do have concerns about the actual significance of the work.

Comments

1. My overarching reaction to the manuscript is that it is more impressive than it is significant. Reviewer 3 shared a similar reaction: ““the authors’ algorithm is cool, but does it sufficiently advance us beyond the capabilities that already exist in large language models, which are available to students and instructors.” Reviewer 3 was perhaps more narrowly focused on the advance of the current methods relative to other large language models or to ChatGPT, but I think the point applies more generally. I might differ from Reviewers 1 and 2, but, to my mind, the empirical results are more of a sanity check than meaningful results on their own. That is, I don’t think these results speak to a theory or debate in the field of cognitive science. Rather, as the authors acknowledge, this is more of a methods paper and I think the empirical results mostly demonstrate that the method is doing what it “should do.” For example, showing that subjects know more about a topic after watching the video or that knowledge about a topic “predicts” performance for similar questions is something that most people would take at face value. Likewise, the fact that questions mapped to fairly specific time points in the lectures is something that almost needs to be true. If the questions are drawn from the lecture content (which they are), then it would be a red flag if the method couldn’t map questions to timepoints in the lecture. Thus, for each of these results, if the model failed to show these effects, it would be a reason to tweak the model, not a meaningful result. (And I think the authors would mostly agree with all of this). In fact, there are quite a lot of degrees of freedom that went into getting results that “make sense.” This is, of course, more evident in the rebuttal than in the actual manuscript.

We agree with the reviewer that the primary focus of our manuscript is methodological rather than empirical: We present a mathematical framework for characterizing learners' knowledge and how it changes with training. Given this goal, the analyses and results we present in our manuscript are primarily intended to validate our approach and demonstrate its capabilities, rather than speak to a particular theory or debate in the field of cognitive science. In our view, an important validation of any model of a complex, real-world process is its ability to recover high-level phenomena we might intuitively "know" should occur, in order to establish a credible basis for any further insights the model might afford. (E.g., if we were developing a model of large body interactions in the field of astrophysics, we might first want to confirm that it can explain a gravity-like attraction between bodies, despite that we already "know" one exists.) In our case, we show that, for example, our approach can map individual quiz questions onto specific, relevant sections of lecture content (Fig. 4; Supp. Tab. 3; Supp. Figs. 3–4) and recover increases in learners' knowledge about each lecture's content after they view it (Fig. 5). The value of these results is not that they would be unobtainable via any other method (e.g., one could qualitatively match questions to lecture content and manually score participants' quiz responses to achieve something similar), but rather in demonstrating (A) that our approach can recover and quantify these patterns *automatically*, and (B) that the latent space in which we embed the lectures and questions to characterize relationships between them captures something meaningful about their underlying conceptual contents.

Following these results, we then present insights that *wouldn't* be obtainable via traditional methods, and which depend on both (A) and (B) being true. For example, we show that we can leverage the automatically discovered conceptual relationships between quiz questions to reliably predict learners' success on specific held-out questions from their performance on specific held-in questions (Fig. 6). We also characterize how participants' knowledge "falls off" with increasing "conceptual distance" from content they know about (Fig. 7) and demonstrate how our approach can be extended to predict knowledge for any arbitrary content expressible by the text embedding model we trained (Fig. 8). In other words, while the initial set of "sanity-check" results are not intended to constitute significant empirical insights in and of themselves, they play an important role in validating the foundation we subsequently build upon to obtain more meaningful insights into learners' knowledge and how it may be characterized via our framework.

2. I think the question is whether the paper does enough, as a methods paper. (Again, I think this is where Reviewer 3's concerns were, as well). As I note above, I think the authors have done enough on the statistics/analyses front to make a compelling case that the method "works." That said, the paper does not introduce a new large language model, nor does it even use the latest models. Thus, I think the value as a methods paper comes down to the specific way in which existing models are leveraged to 'map' knowledge. Ultimately, I am not particularly convinced that there is much practical value in this method. The authors motivate the ideas in terms of teachers that might want to map what students know. Putting aside concerns about the actual feasibility of a teacher using this, the immediate question that came to my mind was: what is the advantage (gain) of this method relative to much simpler approaches. The authors compare their method against a strawman of a single test grade (e.g., 85%). The argument is that a single test grade makes it impossible to assess which specific topics or concepts a student has learned. That's true, of course. But, for most of us that teach classes, it is very common (and almost a necessity) that we structure

exams so that questions are organized around sub-topics. For example, we might deliberately include a certain number of questions per subject area. It is therefore quite trivial to have sub-scores by topic area. While I genuinely loved the visualizations that were presented in the current paper (again, it is beautiful work), I really struggled to see how or whether these visualizations provide more useful information than sub-scores on an exam. Even for the analyses that seek to predict performance on held-out questions, I was left wondering whether this method is any more sensitive than simple conditional probabilities based on sub-scores. For example, if a student answered the first 4 questions on a sub-topic correctly, what is the probability of answering question 5 correctly? What if they only answered 2 of the first 4 questions correctly? For a methods paper, I think that there is somewhat of a burden to demonstrate that the new method being presented outperforms existing methods. And, ultimately, I was not convinced that the current paper does this. Yes, the high resolution coordinate spaces are impressive and they certainly give the impression of being much more nuanced than a sub-score, but it is harder to see the actual application of that. Can higher-resolution feedback be provided to a student or teacher? And would this higher-resolution feedback make a meaningful difference to learning outcomes? In these respects, there is no clear evidence or argument that this new method is superior to much, much simpler alternatives.

The reviewer is correct that, as opposed to developing a new large language model, the goal of our work is to develop a mathematical framework for characterizing knowledge and learning. It is also true that our framework leverages (in part) an older, “simpler” text embedding model (Latent Dirichlet Allocation) rather than the “latest” LLMs. As the reviewer notes above, we contextualize this choice of model in our manuscript, supplement, and responses to previous reviewers’ comments: In brief, while state-of-the-art LLMs are useful for a wide range of applications, there are certain fundamental properties of these models (such as the enormous conceptual breadth of their training corpora) that tend to make them less suitable for our particular use compared to smaller models that can be quickly trained on the specific content for which we want to characterize learners’ knowledge (see our *Discussion* and *Supplementary Results* sections, Supp. Fig. 10, and response to comment 2b in our third response letter).

Regarding the value of our approach to predicting success on held-out questions versus doing so based on conditional probabilities derived from sub-scores: this is precisely what is tested by the “Within-lecture” analyses shown in Figure 6. Here, for each of the three quizzes, we consider only the subset of 5 questions a given participant answered about one of the two lectures viewed in our study. We hold out each of these 5 questions, in turn, and estimate the participant’s knowledge at its embedding coordinate using their performance on the remaining 4 questions. Suppose that of these 4 questions, a participant answered 2 correctly. If, instead of using our knowledge-estimation approach, we simply treated this partial score as their probability of answering the 5th (held-out) question correctly, we would obtain a probability of 0.5. Importantly, however, this probability would be 0.5 regardless of *which* 2 of the first 4 questions the participant answered correctly, or *what* the 5th question actually was. Since this probability would be the same for any potential 5th question, it would necessarily provide no signal by which to distinguish questions this individual is more or less likely to correctly answer.

The greater sensitivity of our knowledge-estimation approach comes from considering the latent conceptual themes probed by the specific questions an individual answered correctly and incorrectly, and how they reflect the conceptual themes probed by the specific question we want to predict their ability to answer. In the middle two rows of Figure 6, we show that, for 5 of the 6 tests we performed using only the subset of quiz questions about a single content area, estimated knowledge was reliably higher for correctly answered held-out questions than for incorrectly answered held-out questions. To perform each of these statistical tests, we fit a generalized linear mixed model to explain participants' success on held-out questions given their estimated knowledge for them (with random effects to account for variation among participants and questions). We then performed a likelihood-ratio test comparing this model to a null model which assumed a given participant's estimated knowledge was the same for all held-out questions—as would be the case if we instead used the simple probabilities described above. Thus we do explicitly demonstrate through statistical analyses that our approach outperforms the existing method the reviewer describes.

More broadly, our point is that a sub-score on exam questions about a particular “sub-topic” is no different from an overall score on a shorter exam with a narrower conceptual focus. Sub-scores might provide some information about “how much” different individuals know about a certain sub-topic, or their different likelihoods of correctly answering any additional question about it, but they afford no information about *what* knowledge an individual has about that sub-topic, or *which* specific questions about it they might be more or less likely to correctly answer. While our “Within-lecture” analyses define “sub-topics” at the scale of the content covered in an 8- to 10-minute lecture video, this would be true of any scale of sub-topic about which one could ask a learner 5 meaningfully different questions—a given section of a lecture might cover a sub-topic of the full lecture, which might cover a sub-topic of a larger academic unit, which might in turn represent a sub-topic of a full-length course, and so on. We would therefore disagree with the characterization of our comparison to a simple test grade (e.g., 85%) as a “strawman.” Manually defining “sub-topics” at any conceptually meaningful level would ultimately entail characterizing learners' knowledge about each of those sub-topics via an analogously impoverished proportion-correct score. And while sub-scores on more specific categorical sub-topics might provide some additional precision over sub-scores on broader sub-topics, an important advantage of our approach is that we model the relationships between to-be-learned concepts *continuously*, allowing us to characterize knowledge at an arbitrarily high level of specificity (e.g., a single point in a high-dimensional space defined by a particular conceptual domain or subdomain of interest).

3. *One way to demonstrate the value of the method would be to actually run a classroom experiment showing that students and/or teachers benefitted from the high-resolution maps compared to some simpler alternative. I realize the authors would argue that this is beyond the scope of the current paper (and I agree), but I think if the scope of the current paper is simply to demonstrate the method without demonstrating the actual value of the method, then it is necessarily a less significant contribution.*

We agree that such a study is beyond the scope of our current paper. However, we do think that our existing analyses, such as those we describe in response to the reviewer’s comment above, demonstrate the value of our approach over common existing measures.

4. Relatedly, Reviewer 3 raises the point that it’s not clear the current model would outperform ChatGPT in the core function of assessing conceptual knowledge. I tend to agree. I do get the authors’ points that their method can track and map learning in a way that ChatGPT isn’t capable of (I agree with this), but, to me, it really comes down to the potential value of the method. This gets back to my question of whether the high resolution maps can actually be translated to some meaningful feedback or intervention. But if the goal is simply to have a user-friendly, powerful method for assessing conceptual knowledge, I think ChatGPT is probably the more appealing option.

We certainly agree that ChatGPT is a powerful tool with a user-friendly interface. Importantly, however, the goal of our manuscript is not to develop a market-ready, consumer-facing product like ChatGPT, nor do we view our framework as an “alternative” to ChatGPT or other LLMs in its core function or purpose. Rather, our approach aims to accomplish the separate and specific goal of constructing a mathematical representation that captures an individual’s conceptual knowledge and how it changes as they learn. As the reviewer notes, ChatGPT (and other LLMs) currently do not maintain or leverage any such representation—and we would thus argue that LLMs themselves are incapable of assessing conceptual knowledge in a rigorous or meaningful way.

Both here and in comment #2, the reviewer raises questions regarding how the knowledge representations we construct might be translated into actionable feedback or interventions for learners. We agree this is an important and exciting direction for future work! Just as additional research effort was required to translate the original transformer architecture introduced by Vaswani et al. (2017; “Attention Is All You Need”) into concrete, user-friendly tools like ChatGPT, the same is true of the nascent computational framework we propose in our manuscript. However, as in many papers that present a novel modeling approach, we provide a basic demonstration of one way this could be accomplished. In Figure 8C, we show that we can recover a distribution of weights over words representing the abstract conceptual content underlying any given point on a knowledge map—e.g., a point where an individual’s estimated knowledge is particularly high or low. Here, for visualization purposes, we display these distributions as word clouds (where each word’s size denotes its relative weight). However, one could easily imagine instead using these sets of weights, along with the knowledge estimates at their corresponding coordinates, as inputs to a separate system that translates these mathematical representations of concepts and knowledge into human-digestible feedback or tangible interventions—such as a large language model. In fact, we had described this very possibility in our prior submission’s *Discussion*:

“Another application for large language models that does not require explicitly modeling the content of individual lectures or questions is to leverage these models’ abilities to generate text. For example, generative text models like ChatGPT [56] and LLaMa [69] are already being used to build a new generation of interactive tutoring systems [e.g., 46]. Unlike the approach we have taken here, these generative text model-based systems do not explicitly model what learners know, or how their

knowledge changes over time with training. One could imagine building a hybrid system that combines the best of both worlds: a large language model that can generate text, combined with a smaller model that can infer what learners know and how their knowledge changes over time. Such a hybrid system could potentially be used to build the next generation of interactive tutoring systems that are able to adapt to learners' needs in real time, and provide more nuanced feedback about what learners know and what they do not know."

In ongoing (unpublished) work, we have already begun developing such hybrid systems that combine our approach for representing a learner's knowledge with a large language model that leverages this representation to suggest additional materials to review and adaptively construct future assessments based on them. At its core, this iterative process of constructing and updating an internal representation of a learner's conceptual knowledge, and drawing on it to personalize their instruction, mirrors how a real human teacher might approach guiding a student one-on-one. As Reviewer 3 noted in their initial review, *"there is a vast literature spanning 2-3 decades on the importance of specific feedback to students."* We think the future development and refinement of such automated systems represents an exciting opportunity to extend this sort of individualized feedback and instruction to students for whom it might not typically be available (e.g., those in online courses without access to a live teacher or tutor). In developing a formal method of representing an individual's conceptual knowledge and how it changes over time, our manuscript presents a "prototype" for an essential component of such a system that is currently absent from modern language models.

In our revised manuscript, we have expanded and refined the above quoted paragraph to further clarify the limitations on our framework's ability to provide concrete, actionable feedback to students in its current form, and highlight this as a direction for future work.



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